A Giant Computer Model Lights Up the Universe's Dark Age

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Imagine the early universe as a dark room with no lights on. Scientists call this time the **cosmic dark age**, and a powerful new computer **simulation** called Lumina is helping them explore it.

Scientists at MIT, Harvard, and other schools built Lumina. It recreates about 2 billion years of cosmic history across a region about 1.6 billion **light-years** wide and shows how the first galaxies formed.

The story starts with the Big Bang. As the universe cooled, **electrons** joined atomic nuclei to form hydrogen. About 380,000 years after the Big Bang, the dark age began, with no stars yet to light it.

Then the first stars and galaxies appeared. Their ultraviolet light knocked **electrons** off hydrogen atoms, a change called **re-ionization**, and the universe grew clearer.

Lumina also follows helium **re-ionization**, a harder change that needs much stronger light. The results suggest feeding **supermassive black holes** supplied it, while stars and galaxies handled the hydrogen.

MIT physics professor Mark Vogelsberger says Lumina covers star and galaxy formation and the **re-ionization** of hydrogen and helium, in a space 100 times bigger than earlier **simulations**.

Lumina builds on an earlier **simulation** called IllustrisTNG, and the team is sharing it so other scientists can check its ideas against telescope observations.

Word Treasure

cosmic dark age -- A long time in the early universe when there were no stars yet to give off light.

simulation -- A computer model that copies how something real works so scientists can study it.

light-years -- A way to measure very long distances in space by how far light travels in one year.

electrons -- Tiny parts of atoms that carry a small electric charge.

re-ionization -- A change in which light knocks electrons off atoms again after they had joined together.

supermassive black holes -- Huge black holes at the centers of galaxies that pull in gas and stars with very strong gravity.

Background you need

The cosmic dark age was a time in the early universe when there were no stars yet, so space was dark.

A computer simulation is a model built by scientists to copy how something works, so they can study it even when they cannot see it directly.

Lumina is a powerful computer model that recreates how the first stars and galaxies formed and how their light changed the gas around them.

Break it down

WHO: Scientists at MIT, Harvard, and other schools, including MIT physics professor Mark Vogelsberger.

WHAT: They built Lumina, a powerful computer simulation that recreates about 2 billion years of cosmic history and shows how the first galaxies formed and how hydrogen and helium were re-ionized.

WHERE: In a simulated region about 1.6 billion light-years wide, a space 100 times bigger than earlier simulations covered.

WHY: To explore the cosmic dark age and let other scientists check its ideas about the early universe against telescope observations.

Why it matters

Just like turning on a light helps you find your way in a dark room, Lumina helps curious kids understand how the first stars lit up the universe we live in today.

Test what you remember

1. What is Lumina?

- (A) A star in the early universe
- (B) A new kind of telescope
- (C) A powerful computer simulation of the early universe
- (D) A planet 1.6 billion light-years away

2. About how much cosmic history does Lumina recreate?

- (A) 100 million years
- (B) 380,000 years
- (C) About 2 billion years
- (D) 1.6 billion years

3. When did the cosmic dark age begin?

- (A) 2 billion years after the Big Bang
- (B) When Lumina was built
- (C) When the first black holes formed
- (D) About 380,000 years after the Big Bang

4. What happened to hydrogen atoms during re-ionization?

- (A) They joined together to form stars
- (B) Ultraviolet light knocked electrons off them
- (C) They disappeared completely
- (D) They froze into ice

5. According to Lumina's results, what supplied the strong light needed for helium re-ionization?

- (A) Feeding supermassive black holes
- (B) The first stars alone
- (C) Light from the Big Bang itself
- (D) Telescopes on Earth

6. What earlier simulation does Lumina build on?

- (A) IllustrisTNG
- (B) The cosmic dark age
- (C) The Hubble telescope
- (D) The Big Bang

Answer key (try the quiz first!): 1: C 2: C 3: D 4: B 5: A 6: A

Questions to think about

1. MIT physics professor Mark Vogelsberger: Says Lumina covers star and galaxy formation and the re-ionization of hydrogen and helium in a space 100 times bigger than earlier simulations.

MY THOUGHTS (at least 20 words):